Weekly Meeting

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Estimating LoRa Range Between Buildings in Halifax

- My initial plan was to only use data collected from previous tests to estimate, on average, what range LoRa can achieve from node to node within an urban environment
  - Using Meshtastic's LONG_FAST setting.
- However, I quickly came to realize that I did not have sufficient data, and needed to collect more.
  - I had plenty of data but only for one route.
- I decided go to other buildings and send packets every 20 seconds between different destinations (zero-hop)
  - This turned out to be highly time-inefficient

The Solution

- I decided that instead of moving to different buildings every time I wanted to test a new path, I instead decided to use traceroutes and capture hop data using traceroute logs.
- From each node, I will periodically send a traceroute to a different node, observing each hop taken on the path, noting various characteristics about such path including:
  - SNR
  - Distance
  - Elevation
  - Reliability?
- Major Issue: Survivor Bias

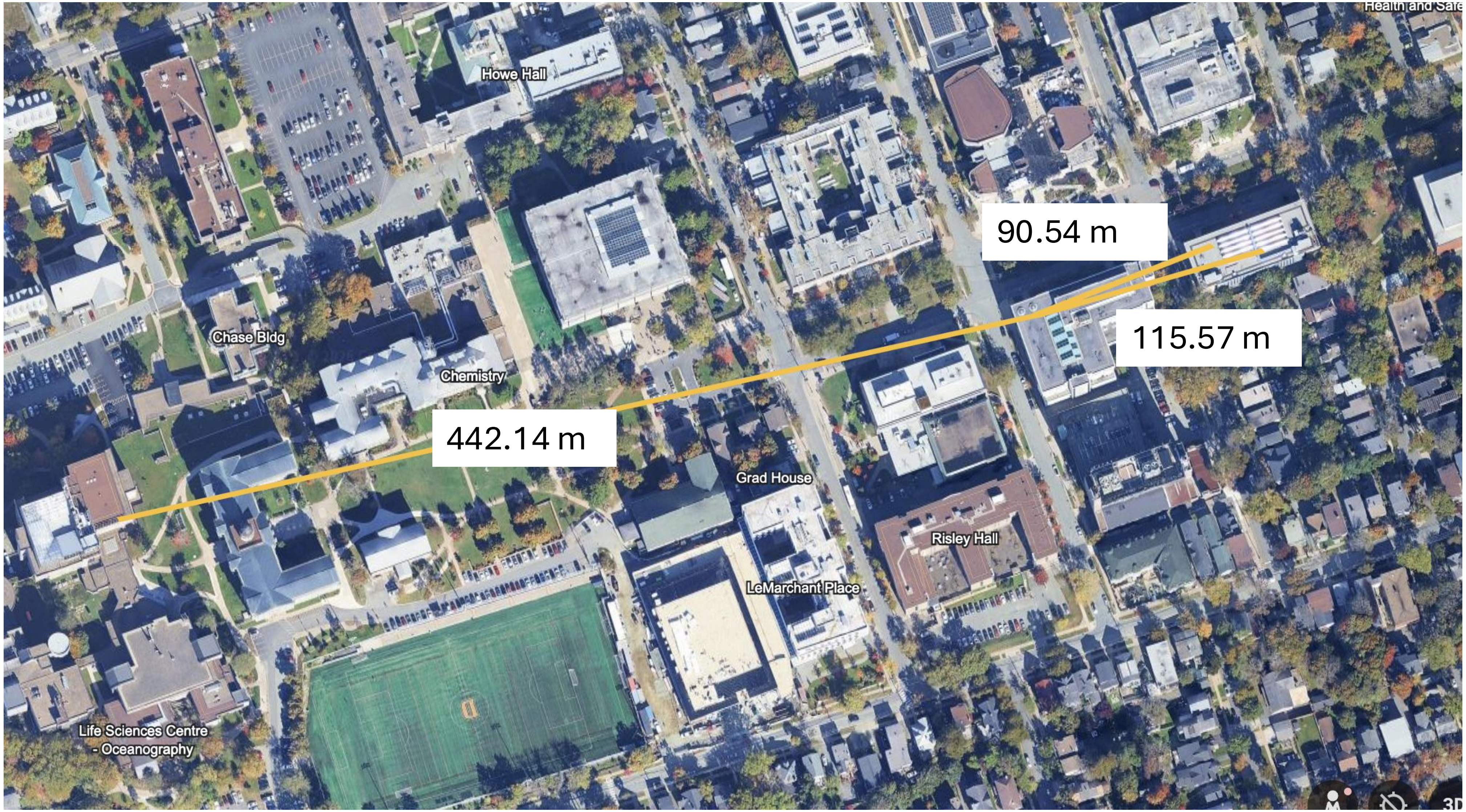
Caveat

- The only issue with the traceroute method is that I cannot track what nodes did successfully receive the traceroute on a failed attempt.
  - This makes it impossible to give a distinct delivery rate metric for each hop.
  - Makes it impossible to track where exactly the traceroute failed
- Instead, I will have to rely on other metrics including:
  - Hop usage among successful traces
  - Observed successful traversal per attempt
- I believe that the time saved using the traceroute method is worth giving up an accurate delivery rate metric.
- A high observed successful traversal per attempt for one traceroute destination should be weighted much stronger than a low rate.

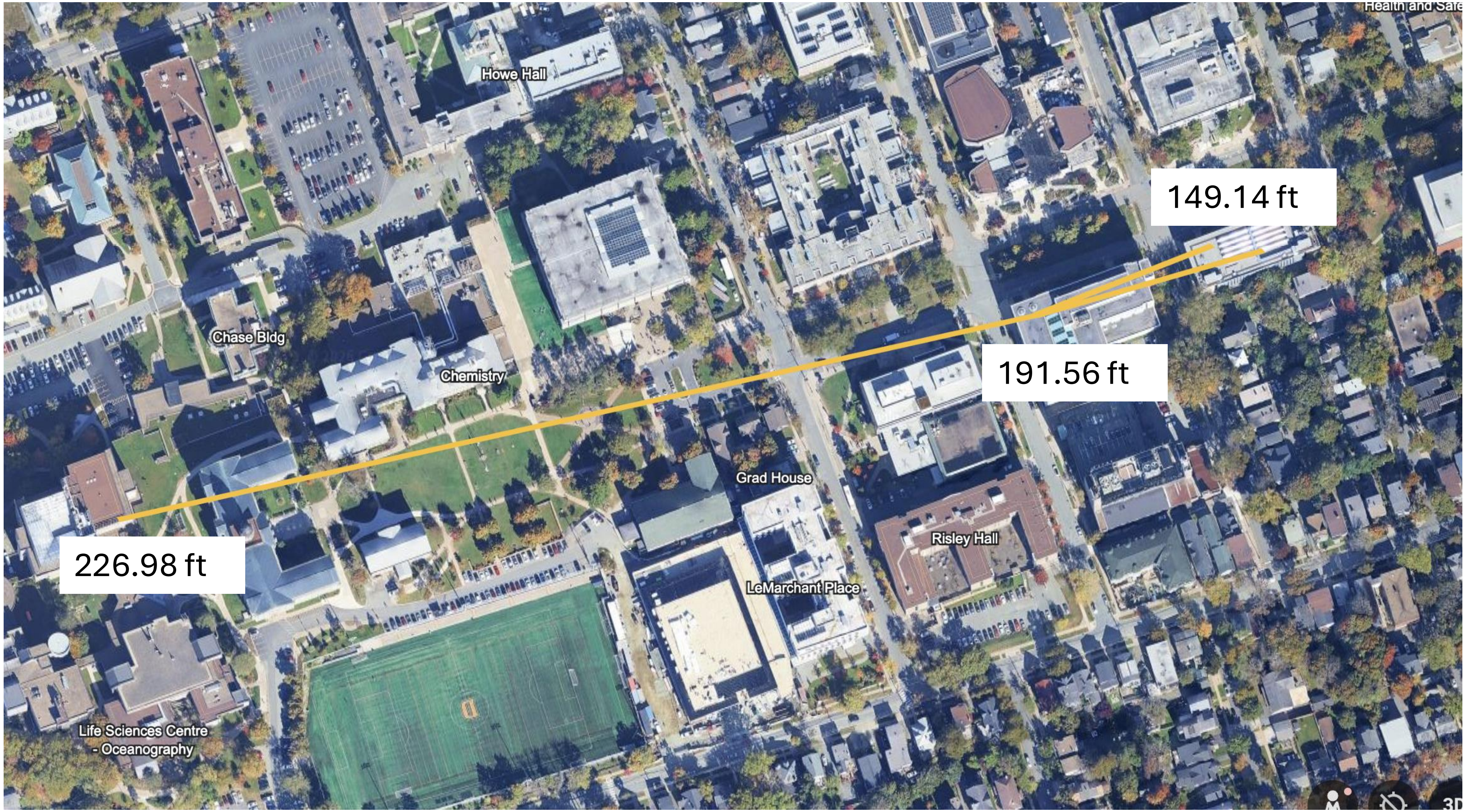
From Node (hop)	To Node (hop)	**Distance (m)**	**Elevation Diff.* (ft)**	Average SNR	Total Occurrences	Occurrences to b0cc (Rowe 5th)	Hop Usage Among Successful Traces (%)	Observed successful traversal per attempt (%)
GB Playground	GB window	20	0	11.25	1	0	N/A	N/A
GB window	GB Playground	20	0	12.05	5	2	N/A	N/A
GB Playground	Rowe	111.57	+42.42	-14.0390625	160	116	100	78.91156463
Rowe	GB Playground	111.57	-42.42	-8.729166667	156	114	98.27586207	77.55102041
Rowe	LSC Biology	442.14	+35.42	-14.23295455	44	0	0	0
LSC Biology	Rowe	442.14	-35.42	-19.27840909	44	0	0	0
Rowe	GB window	90.54	-42.42	6.375	4	2	1.724137931	1.360544218

From Node (hop)	To Node (hop)	**Distance (m)**	**Elevation Diff.* (ft)**	Average SNR	Total Occurrences	Occurrences to b008 (LSC Biology 8th)	Hop Usage Among Successful Traces (%)	Observed successful traversal per attempt (%)
GB Playground	GB window	20	0	11.25	1	0	N/A	N/A
GB window	GB Playground	20	0	12.05	5	2	N/A	N/A
GB Playground	Rowe	111.57	+42.42	-14.0390625	160	44	100	30.1369863
Rowe	GB Playground	111.57	-42.42	-8.729166667	156	42	95.45454545	28.76712329
Rowe	LSC Biology	442.14	+35.42	-14.23295455	44	44	100	30.1369863
LSC Biology	Rowe	442.14	-35.42	-19.27840909	44	44	100	30.1369863
Rowe	GB window	90.54	-42.42	6.375	4	2	4.545454545	1.369863014

Distance

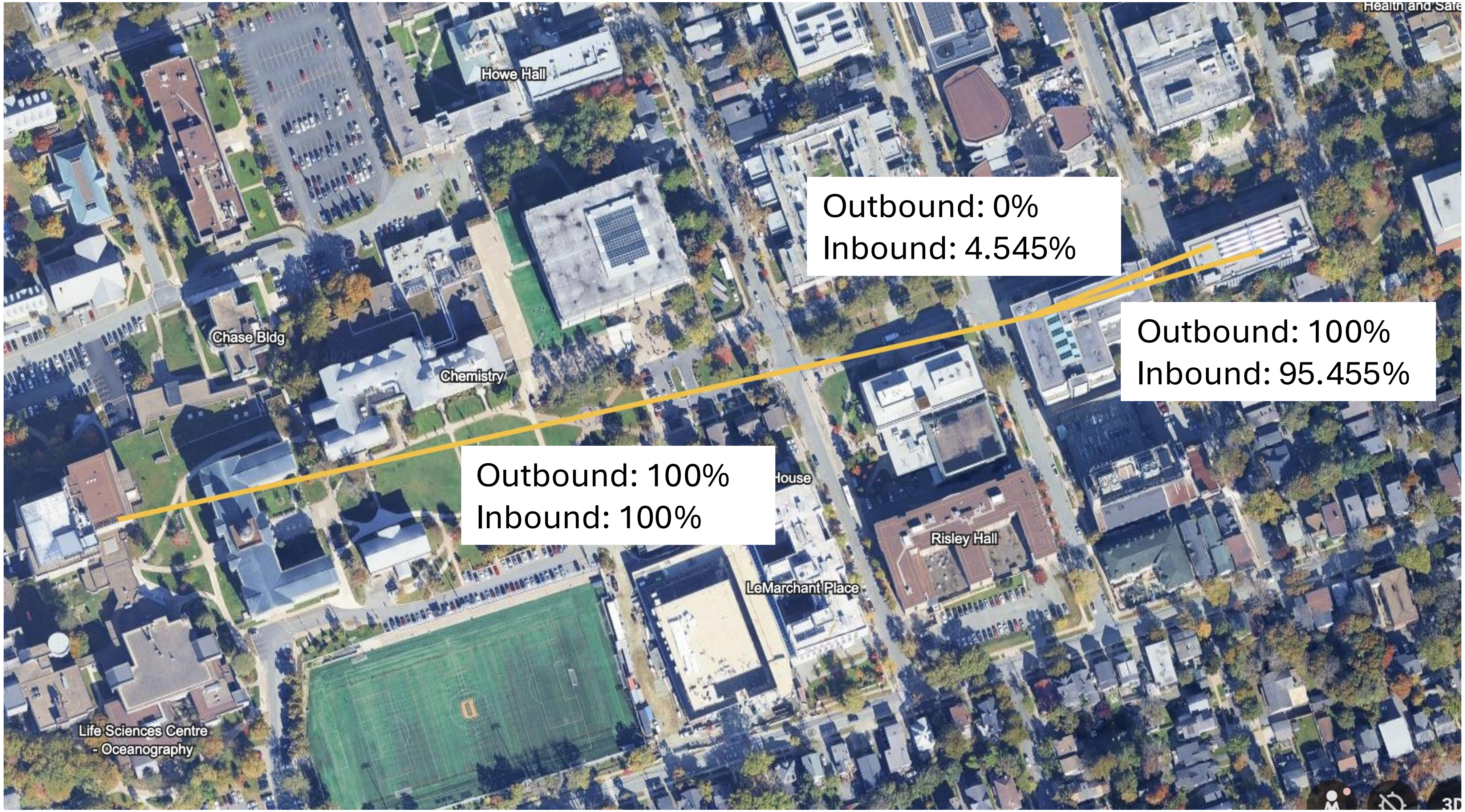

Howe Hall
Chase Bldg
Chemistry
Grad House
Risley Hall
LeMarchant Place
Life Sciences Centre
- Oceanography
90.54 m
115.57 m
442.14 m

Approximate Elevation

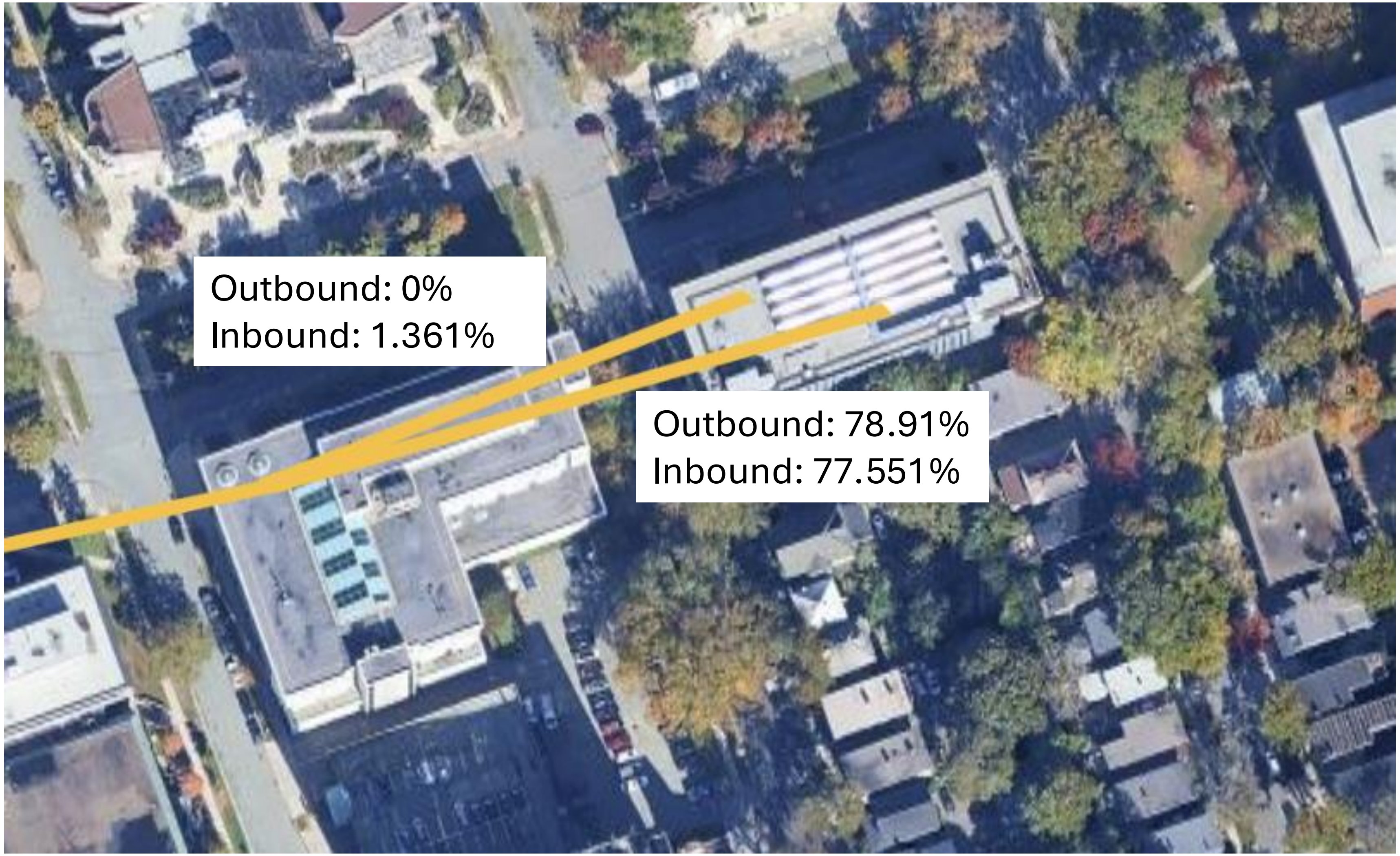


Hop Usage Among Successful Traces (%) - Rowe

Hop Usage Among Successful Traces (%) – LSC



Observed Successful Traversal per Attempt (%) - Rowe



Observed Successful Traversal per Attempt (%) – LSC